

Ang Cao

Research Scientist, Google DeepMind

ancao@umich.edu • caoang@google.com • caoang327.github.io • Google Scholar

Research Appointments

Research Scientist, Google DeepMind

Dec. 2025–Present

Currently focusing on developing agentic capabilities for Gemini centered on self-correction and self-improvement, with applications in 2D/3D code generation, autonomous research for model pretraining, and robotics.

Selected projects (lead or core contributor; most recent first):

- Autonomous pretraining research at scale.
- Agentic robotics through recursive co-improvement of agents and environments.
- Gemini 2D/3D agentic de-rendering via reinforcement learning for code generation with visual feedback.
- Gemini-Omni 3D/4D world modeling.

Research Scientist Intern, Meta FAIR, Menlo Park

May 2024–Dec. 2024

Mentors Sasha Sax and Franziska Meier.

Focus Proposed LIFT-GS, a 3D vision-language grounding pipeline using render-supervised learning from 2D foundation models, demonstrating strong data-scaling properties.

Research Scientist Intern, Meta GenAI, London

May 2023–Nov. 2023

Mentors David Novotny, Andrea Vedaldi, and Natalia Neverova.

Focus Designed a highly scalable GPU operator for neural-field rendering (Lightplane), achieving up to a 10,000× reduction in memory usage compared with existing implementations, and developed feed-forward 3D reconstruction and generation methods.

Education

Ph.D., Computer Science and Engineering, University of Michigan

2020–2025 | GPA 4.0/4.0

Thesis: *Toward Spatial Intelligence via Data and Compute Efficiency*. Advisor: Justin Johnson.

Committee: Andrew Owens, Jeong Joon Park, Andrea Vedaldi, and Stella Yu.

M.S., Electrical and Computer Engineering, University of Michigan

2018–2020 | GPA 4.0/4.0

B.S., Electrical Engineering, Wuhan University

2014–2018 | GPA 3.9/4.0

Publications

* Equal contribution. † Mentored project. Citation record on Google Scholar.

[1] **Generative Point Tracking and Forecasting**

Xuanchen Lu, **Ang Cao**[†], Chao Feng, Andrew Owens

CVPR 2026 | Project

[2] **MoE3D: A Mixture-of-Experts Module for 3D Reconstruction**

Zichen Wang, **Ang Cao**[†], Liam J. Wang, Jeong Joon Park

arXiv 2026

[3] **Probing Visual Language Priors in VLMs**

Tiange Luo^{*}, **Ang Cao**^{*}, Gunhee Lee, Justin Johnson, Honglak Lee

ICML 2025 | Project

[4] **From Thousands to Billions: 3D Visual Language Grounding via Render-Supervised Distillation from 2D VLMs**

Ang Cao, Sergio Arnaud, Oleksandr Maksymets, . . . , Alexander Sax

ICML 2025 | Project

[5] **LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D**

Paul McVay, Sergio Arnaud, Ada Martin, . . . , **Ang Cao**, . . . , Franziska Meier

ICML 2025 | Project

Publications (continued)

- [6] **Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass**
Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, **Ang Cao**, Joyce Chai, Franziska Meier, Matt Feiszli
CVPR 2025 | Project Code
- [7] **Lightplane: Highly-Scalable Components for Neural 3D Fields**
Ang Cao, Justin Johnson, Andrea Vedaldi, David Novotny
3DV 2025 | Project Code
- [8] **Meta 3D Gen**
Raphael Bensadoun, Tom Monnier, Yanir Kleiman, . . . , **Ang Cao**, . . . , Andrea Vedaldi
Technical report, 2024 | Project
- [9] **DreamGaussian4D: Generative 4D Gaussian Splatting**
Jiawei Ren*, Liang Pan*, Jiaxiang Tang, Chi Zhang, **Ang Cao**, Gang Zeng, Ziwei Liu
arXiv 2023 | Project
- [10] **Text2Room: Extracting Textured 3D Meshes from 2D Text-to-Image Models**
Lukas Höllein*, **Ang Cao***, Andrew Owens, Justin Johnson, Matthias Nießner
ICCV 2023 Oral | Project Code
- [11] **HexPlane: A Fast Representation for Dynamic Scenes**
Ang Cao, Justin Johnson
CVPR 2023 | Project Code
- [12] **FWD: Real-Time Novel View Synthesis with Forward Warping and Depth**
Ang Cao, Chris Rockwell, Justin Johnson
CVPR 2022 | Project Code
- [13] **Inverting and Understanding Object Detectors**
Ang Cao, Justin Johnson
Technical report, 2021 | Paper Code
- [14] **Unified Signal Compression Using Generative Adversarial Networks**
Bowen Liu*, **Ang Cao***, Hun-Seok Kim
ICASSP 2020 Oral | Paper

Honors and Awards

Outstanding Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition	2026
Rackham Conference Travel Grant, University of Michigan	2023
Rollin M. Gerstacker Foundation Fellowship, University of Michigan	2020
Outstanding Graduate of Wuhan University	2018
National Scholarship, Ministry of Education of China	2016, 2017
Meritorious Winner, Mathematical Contest in Modeling	2017
National Undergraduate Innovation Foundation, Ministry of Education of China	2016

Professional Service

Conference reviewer: CVPR (since 2022), NeurIPS (since 2023), ICLR (since 2023), ECCV (since 2022), ICCV (since 2023), SIGGRAPH, and SIGGRAPH Asia.

Journal reviewer: IEEE TPAMI and IEEE TVCG.

Teaching

Head GSI / Teaching Assistant, EECS 498/598: Computer Graphics and Generative Models, University of Michigan, Fall 2024

Invited Talks

“Toward Building a Realistic Digital 3D World.” NVIDIA Spatial Intelligence Lab; Meta FAIR; Google DeepMind; Meta Reality Labs; Stanford Vision and Learning Lab, 2025